2. Action of carbondisulphide with chlorine gas

Carbon disulphide reacts with chlorine gas in the presence of anhydrous $AlCl_3$ as catalyst giving carbon tetrachloride

$$\underset{\text{carbon disulphide}}{CS_2} + 3Cl_2 \xrightarrow[AlCl_3]{\text{anhydrous}} \underset{\text{carbon tetrachloride}}{CCl_4} + S_2Cl_2$$

Physical properties

(i) Carbon tetrachloride is a colourless liquid with its specific smell

(ii) It is insoluble in water and soluble in organic solvents

Chemical properties

(i) Hydrolysis

Carbon tetrachloride reacts with hot water or with hot water vapour producing the poisonous gas, phosgene.

$$\underset{\text{carbon tetrachloride}}{CCl_4} + H_2O(g) \xrightarrow{\Delta} \underset{\text{Phosgene}}{COCl_2} + 2HCl$$

(ii) Reduction

Carbon tetrachloride is reduced by iron powder in dilute HCl medium to form chloroform

$$\underset{\text{carbon tetrachloride}}{CCl_4} + 2(H) \xrightarrow{Fe/HCl} \underset{\text{chloroform}}{CHCl_3} + HCl$$

14.6.4 Freons (CFC)

The chloro fluoro derivatives of methane and ethane are called freons.

Nomenclature

Freon is represented as Freon-cba

Where c = number of carbon atoms – 1

b = number of hydrogen atoms + 1

a = total number of fluorine atoms

Example

Formula	C–1	H+1	F	Name
$CFCl_3$	1–1=0	0+1=1	1	Freon–11
CF_2Cl_2	1–1=0	0+1=1	2	Freon–12
$C_2F_2Cl_4$	2–1=1	0+1=1	2	Freon–112
$C_2F_3Cl_3$	2–1=1	0+1=1	3	Freon-113

Freon – 12 is prepared by the action of hydrogen fluoride on carbon tetrachloride in the presence of catalylic amount of antimony pentachloride. This is called swartz reaction.

$$\underset{\text{Carbon tetrachloride}}{CCl_4} + 2HF \xrightarrow{Sbcl_5} 2HCl + \underset{\text{Freon - 12}}{CCl_2F_2}$$

Physical properties

Freons are highly stable, unreactive, non corrosive, non toxic, easily liquefiable gases.

Uses:

(i) Freons are a used as refrigerants in refrigerators and air conditioners.

(ii) It is used as a propellant for aerosols and foams

(iii) It is used as propellant for foams to spray out deodorants, shaving creams, and insecticides.

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